ServiceNow Certified Application Developer

SmartInternz

Project Title

ServiceNow-Based Employee Laptop Request Application

Platform	ServiceNow
Application Area	Service Catalog and Flow Designer
Primary Catalog Item	Laptop Request
Backend Table	Laptop Request Table
Automation	Flow Designer
Document Type	Project Implementation Report

Name: BODAPATI PRAMEELA
College: CHALAPATHI INSTITUTE OF ENGINEERING AND TECHNOLOGY
Email: y23csm007@chalapathiengg.ac.in

Milestone 1 — Update Set Creation & Project Initialization

1. **Navigate:** All → System Update Sets → Local Update Sets.
2. Click **New** and create an Update Set named HR Onboarding Application.
3. Set the **State** to **In Progress**.
4. Click **Make This My Current Set**.
5. Start developing the project components such as **tables, fields, forms, business rules**.
6. Verify that all changes are captured in the **Customer Updates** section.

Milestone 2 — Custom Application Creation

1. Navigate to **All → System Applications → My Company Applications**.
2. Click **New** and its named as Project application.
3. Define the **Application Scope** and access settings.
4. Verify that all project components are created within the application scope.

Milestone 3 — Table Design & Data Model Configuration

1. Created the **Laptop Request Form** custom table.
2. Added **Auto-number, Reference, and Choice** fields.
3. Configured field validations and data consistency.

Milestone 4 — Request Logic & Business Rules Setup

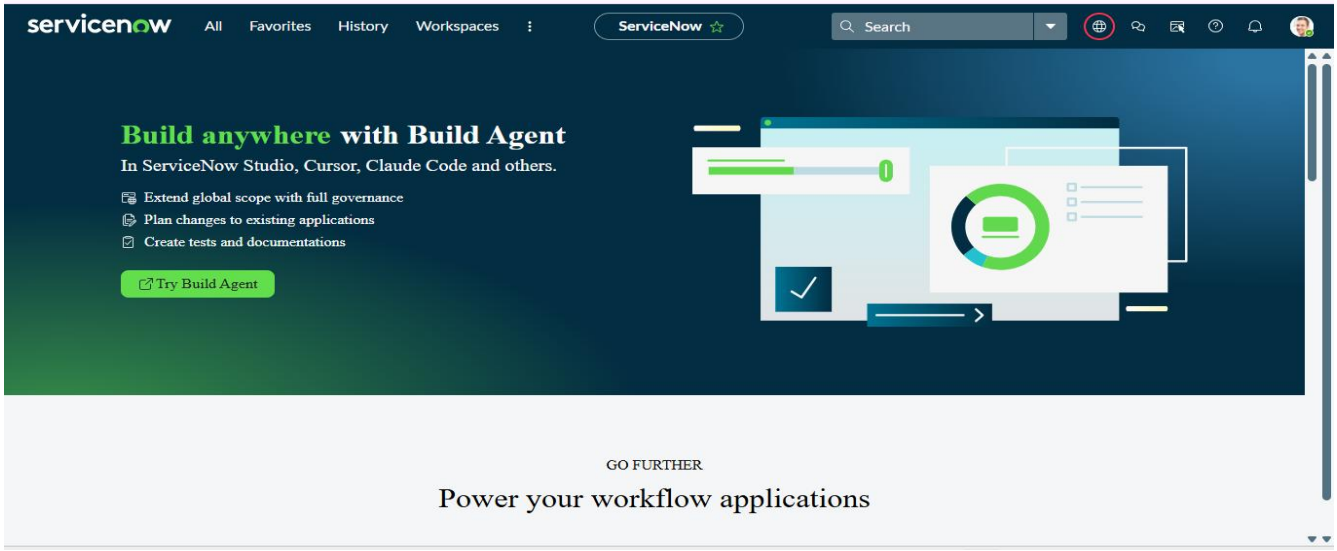
1. Created Business Rules for request processing.
2. Restricted employees to **one laptop request per year**.
3. Added validations and duplicate-request error handling.

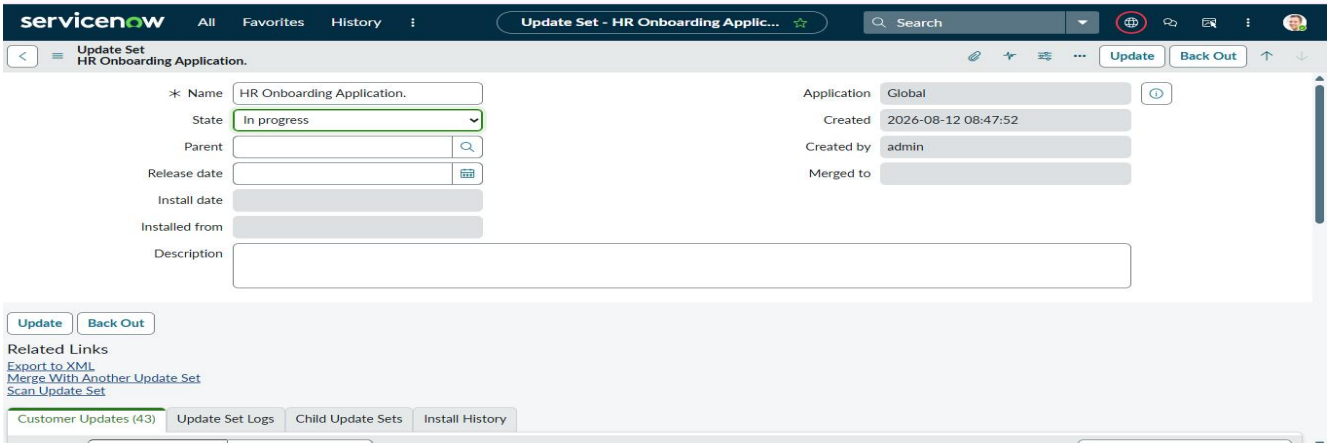
Milestone 5 — Validation, Testing & Readiness for Automation

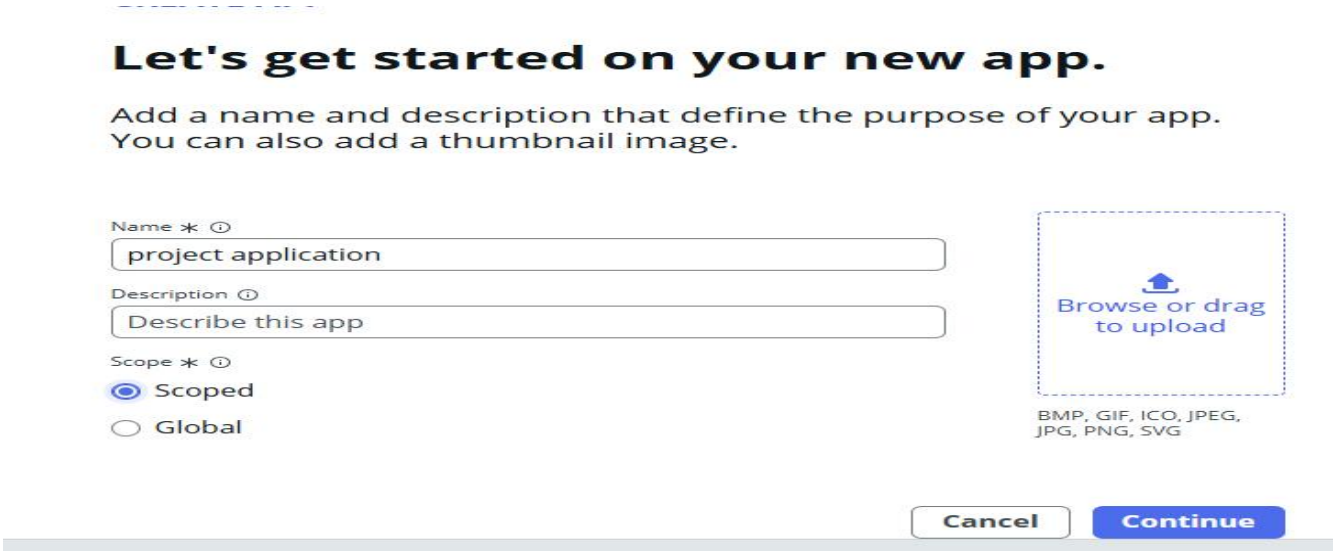
1. Tested the complete **laptop request creation process**.
2. Verified field behavior and choice values.
3. Confirmed all changes were captured in the **Update Set**.

Update Set & Application Creation.

1. **Navigate:** All → **System Update Sets → Local Update Sets**.
2. Click **New** → enter the Update Set name **HR Onboarding Application**.
3. Set **State: In Progress**, add the required details, and click **Submit**.
4. Open the Update Set and click **Make Current**.
5. **Navigate:** All → **ServiceNow Studio**.
6. Click **Create Application** → enter **Project Application**.
7. Select **Scoped Application** → click **Continue**.
8. In the application, go to **Data → Tables** and create the required custom table with the project fields.

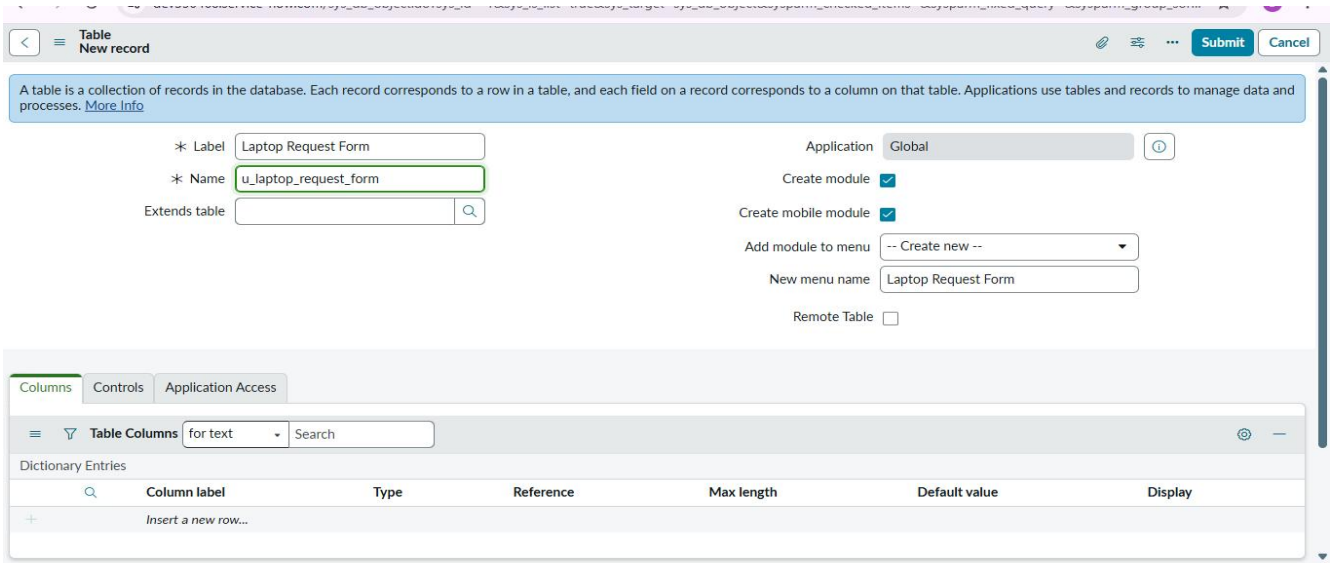


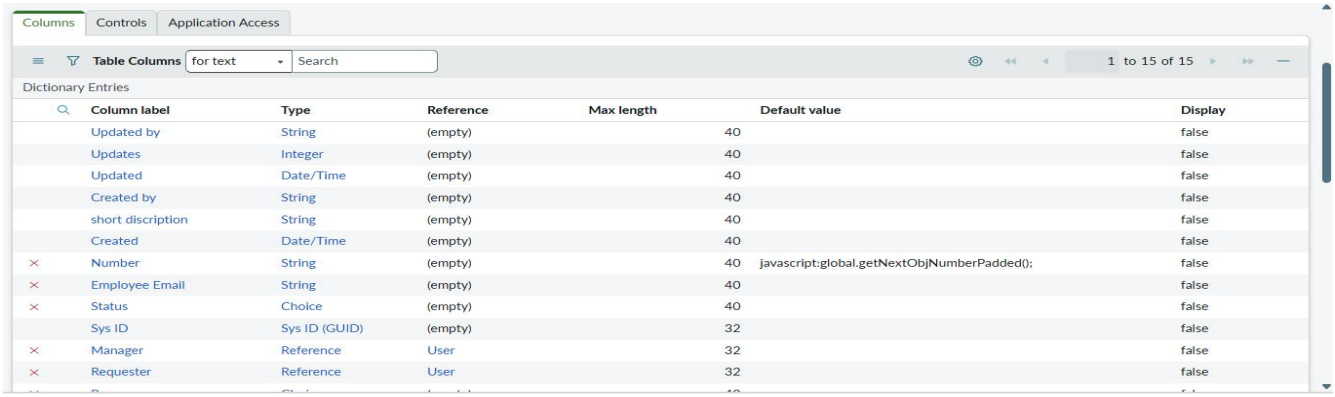




Laptop Request Form Table Creation

1. **Navigation:** ServiceNow Studio → **Project Application → Data → Tables**.
2. Click **New** to create a new table.
3. Enter the table name as **Laptop Request Form**.
4. Configure the required table settings and click **Submit/Save**.
5. Open the **Laptop Request Form** table and add the required fields.
6. Configure fields such as **Employee, Request Type, Reason, Status, Request Date, and Laptop Details**.
7. Save the fields and verify that they are displayed correctly on the table form.

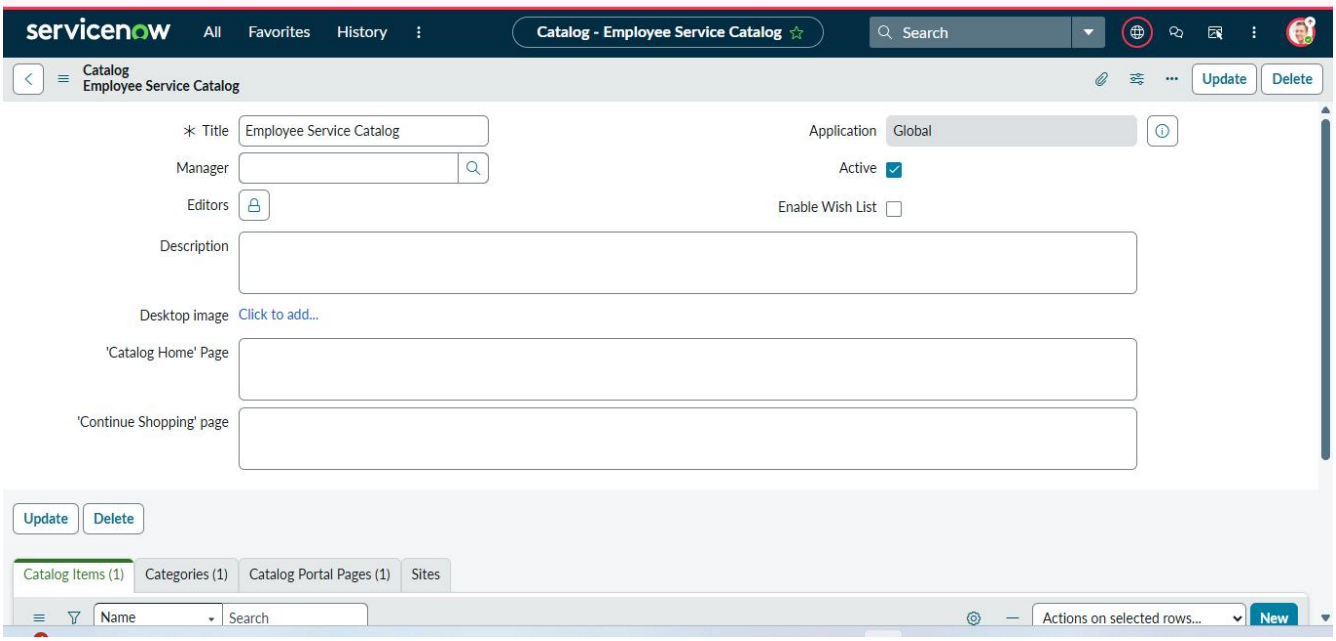




Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
short discription	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Employee Email	String	(empty)	40		false
Status	Choice	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Manager	Reference	User	32		false
Requester	Reference	User	32		false

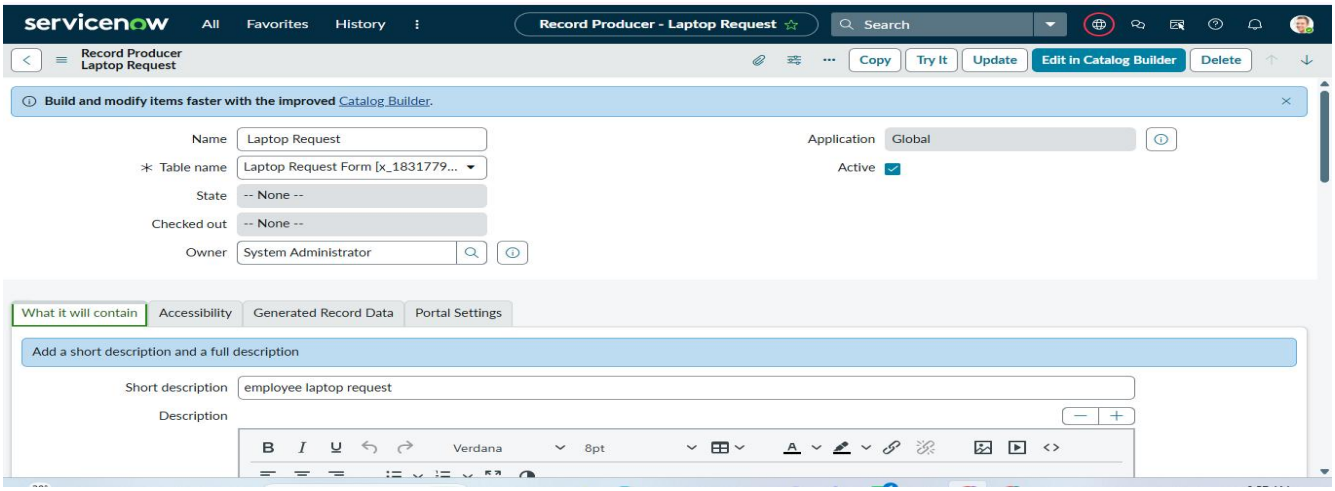
Service Catalog Creation.

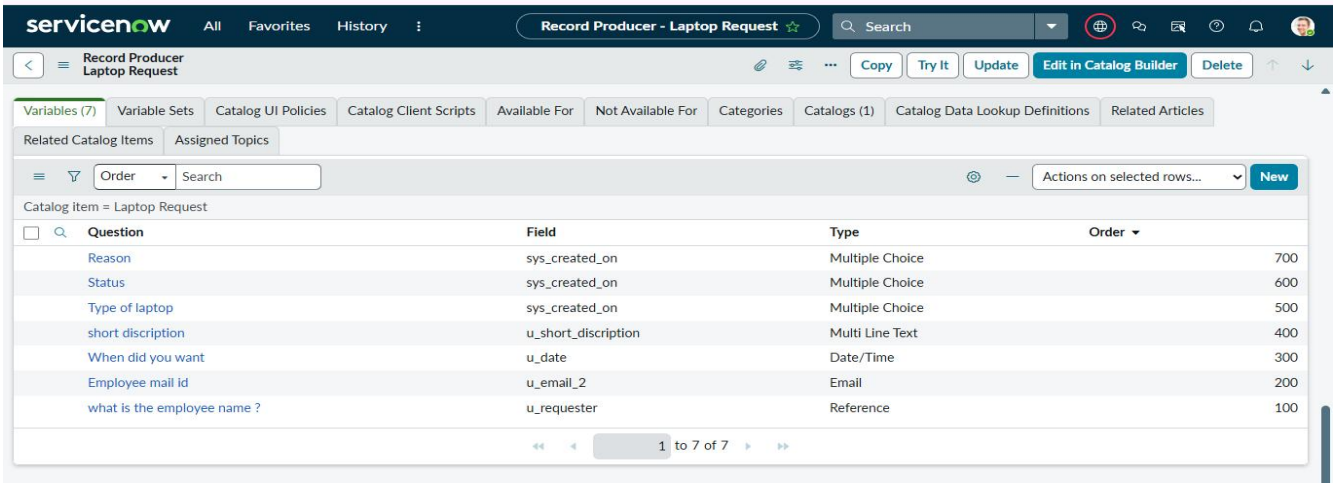
1. **Navigate:** Service Catalog → **Catalog Definitions → Maintain Items**.
2. Create a Catalog Item named **Employee Service Catalog** and select **Service** as the category.
3. Add the **Laptop Request** Record Producer to the Employee Service Catalog.



Record Producers creation

1. **Navigate:** Service Catalog → **Catalog Definitions → Record Producers**.
2. Click **New** and create a Record Producer named **Laptop Request**.
3. Select **Laptop Request Form** as the target table.
4. Add variables matching the required table fields and map them to the corresponding fields.
5. Save the Record Producer.





Flow Designer Automation

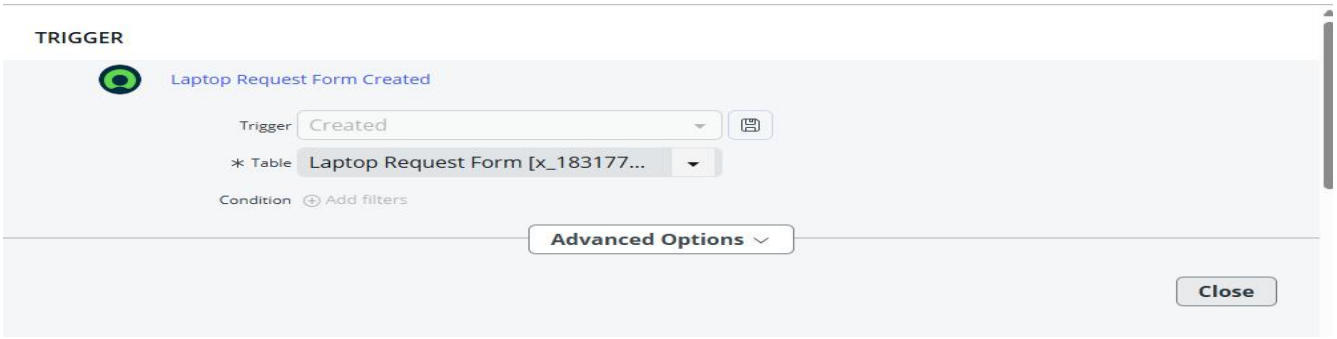
Purpose: Automate the complete laptop request approval process and reduce manual work.

Trigger:

- **Navigate:** Flow Designer → **New Flow**
- Select **Record Trigger**.
- **Table:** Laptop Request Form.
- **When:** Record is **Created**.

Process:

1. Automatically send the request for **Manager Approval**.
2. Update the request **Status** based on approval or rejection.
3. Send **Email Notifications** at each stage.
4. Complete the process automatically without manual intervention.

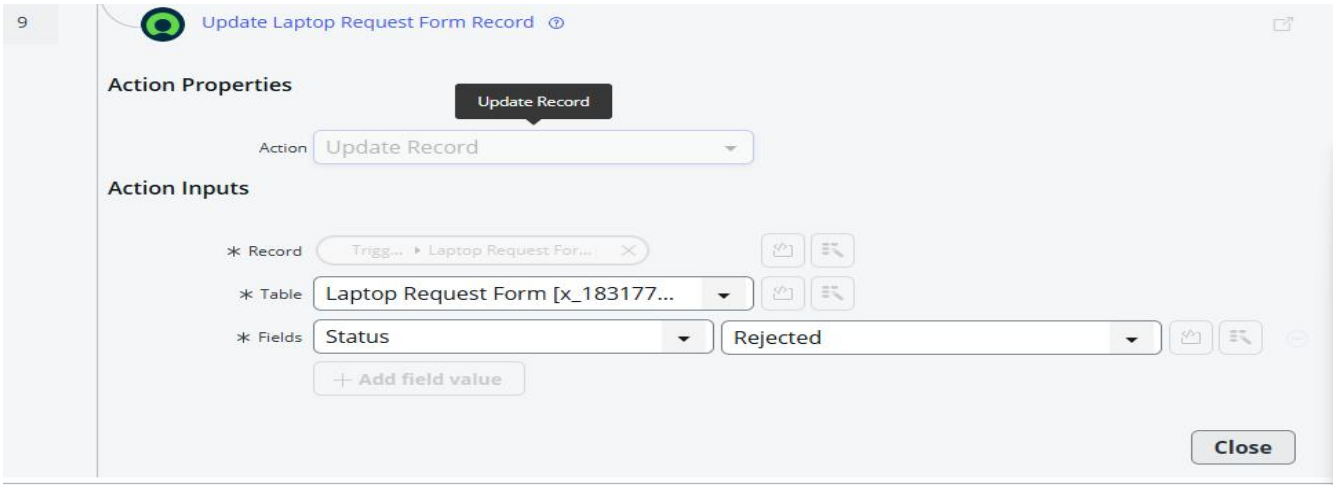




Action — Update Record

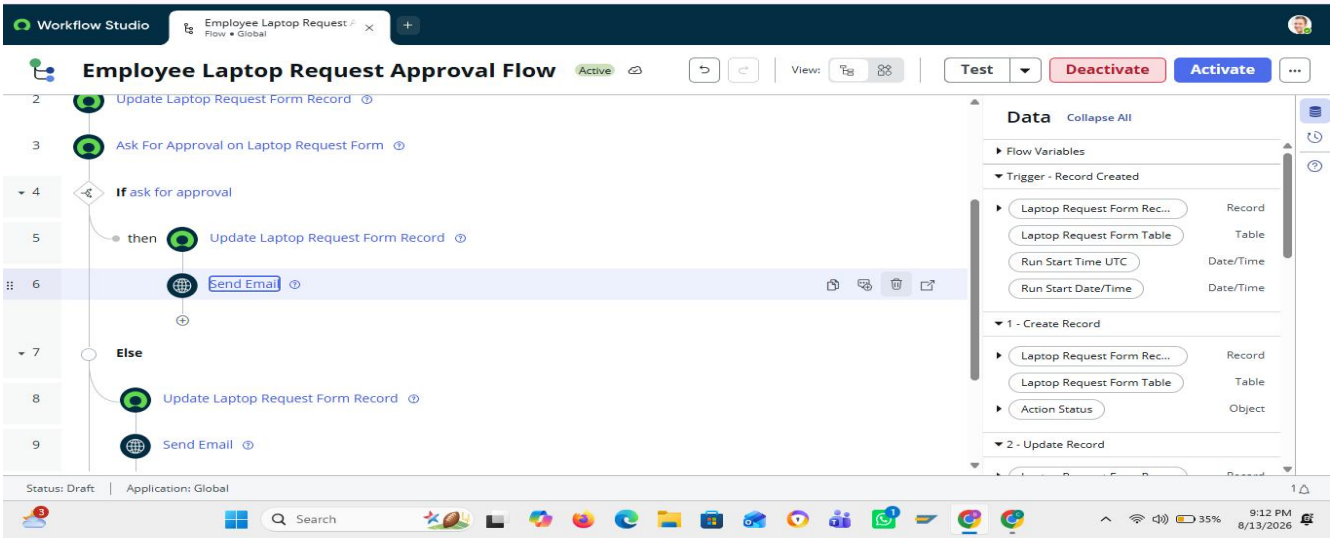
Navigation: Flow Designer → **Actions → Update Record**

- Select the **Laptop Request Form** record.
- Set **Status** to **Pending Approval**.
- This automatically assigns the initial status to every new request.
- It avoids manual status updates and keeps the process consistent.



Approval, If/Else & Email Automation

1. **Navigate:** Flow Designer → **Actions → Ask for Approval**.
2. Select the **Laptop Request Form** record and send it to the **Manager** for approval.
3. Add **If/Else** condition to check the approval result.
4. **If Approved:** Update status to **Approved** and send an approval email.
5. **If Rejected:** Update status to **Rejected** and send a rejection email.
6. **Navigate:** Flow Designer → **Actions → Send Email** to configure the notifications.

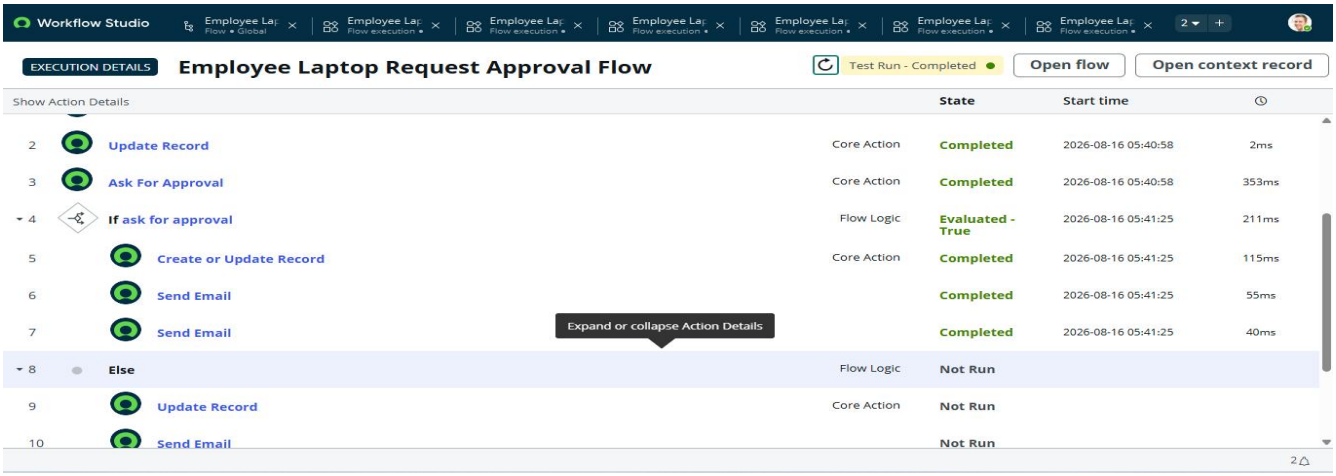


If Request is Approved — Flow

1. **Navigate:** Flow Designer → **Flow Logic → If**.
2. Set the condition: **Approval State → Approved**.
3. **Then:** Use **Update Record** → Set **Status = Approved**.
4. Add **Send Email** → notify the employee that the laptop request has been approved.
5. Save and **Activate** the flow.

If the request is approved then it excutes only IF block this look like below

And also email should be send that is :

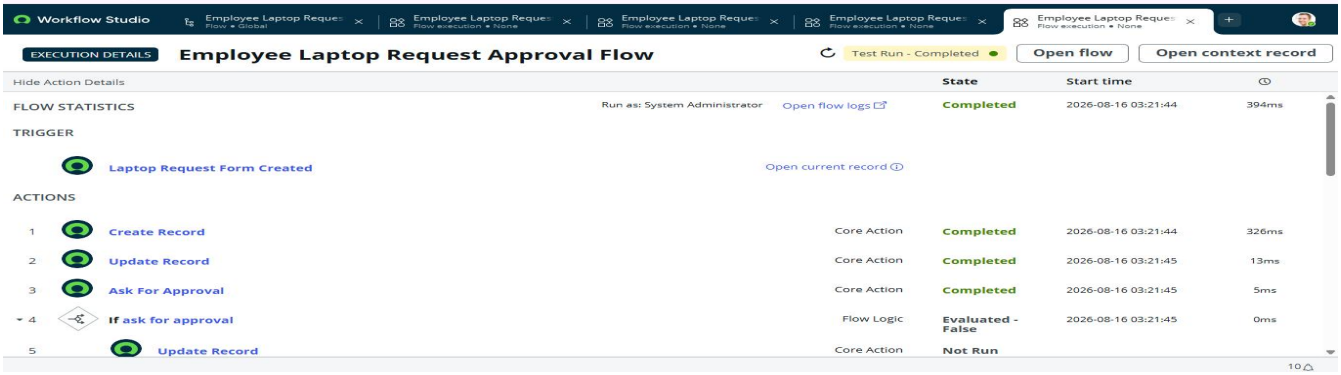


Approved email look like this :



If Request is Rejected — Flow

1. **Navigate:** Flow Designer → **Flow Logic → If**.
2. Set condition: **Approval State → Rejected**.
3. **Then:** Use **Update Record** → Set **Status = Rejected**.
4. Add **Send Email** → notify the employee about the rejection.
5. Save and **Activate** the flow.



It excutes else block that look like this :



The rejected email look loke this .

Final Flow

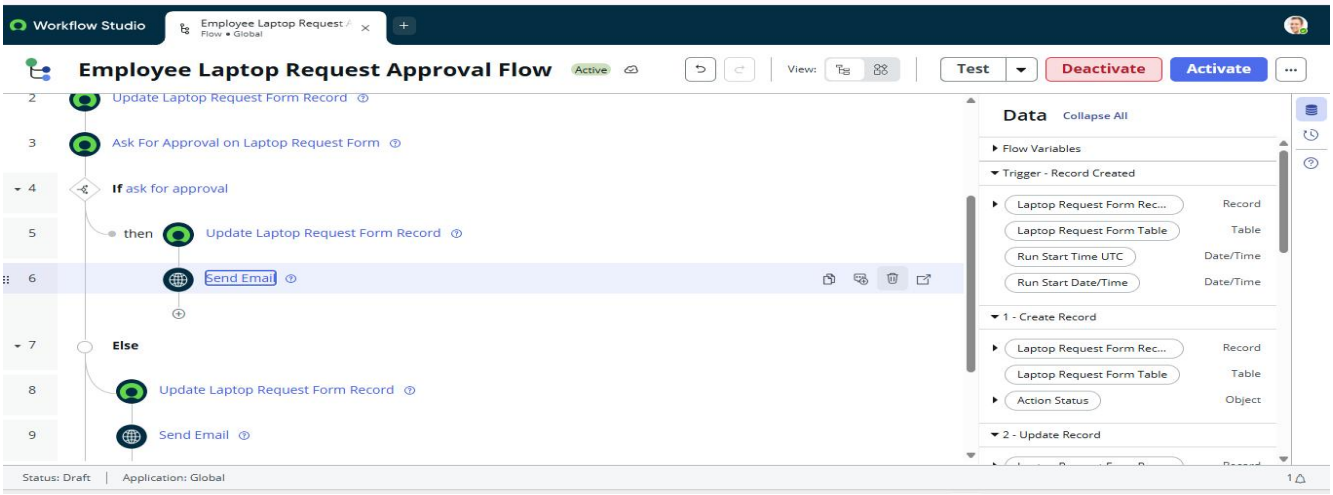
Navigate: Flow Designer → **Laptop Request Flow**

Flow Process:

Request Created → Status: Pending Approval → Manager Approval → If/Else

- **Approved →** Status: Approved → Send Approval Email → End
- **Rejected →** Status: Rejected → Send Rejection Email → End

This completes the automated **Laptop Request Management approval process**



Business Rule – One Laptop Request Per Year

Purpose:
To ensure an employee can submit **only one laptop request per calendar year** and prevent duplicate requests at the backend level.

Navigation:
All → System Definition → Business Rules → New

Configuration:

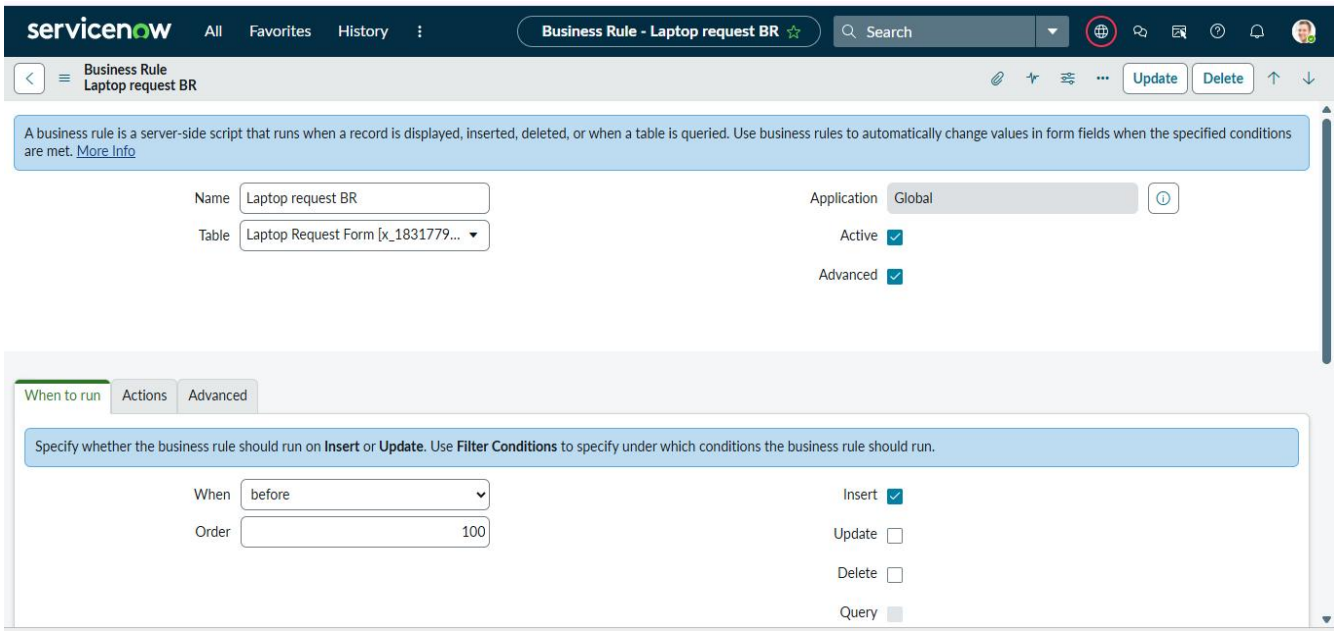
1. **Name:** Laptop request BR
2. **Table:** Laptop Request form
3. Enable **Advanced**
4. **When:** Before
5. Enable **Insert**
6. Go to the **Advanced** section.
7. Add a script that checks whether the requester already has a laptop request in the **current calendar year**.
8. If a request exists, stop the new record and display an error message.

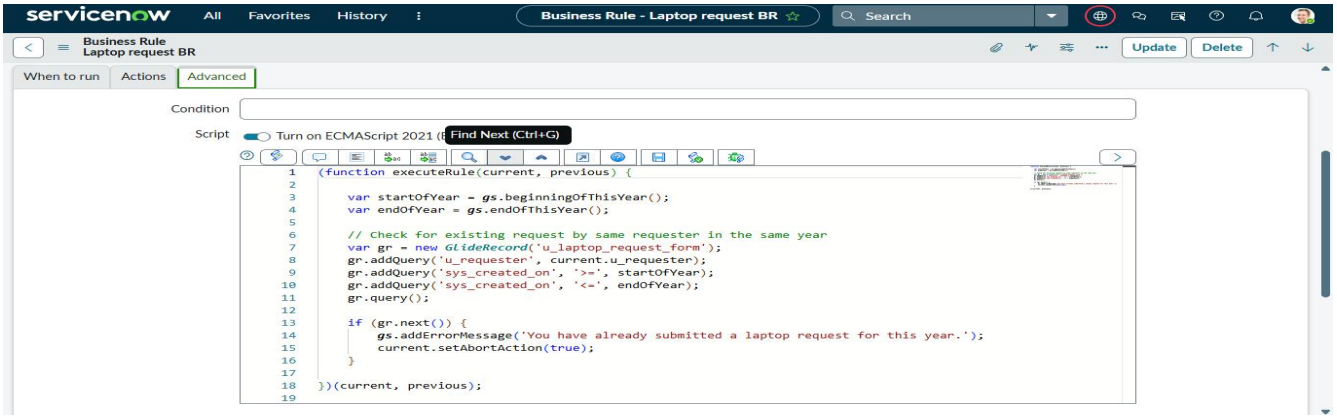
Business Logic – Short Flow

New Laptop Request → Check Requester → Check Current Year → Existing Request?

- **No:** Allow the request to be submitted.
- **Yes:** Block the request and show an error message such as **"You have already submitted a laptop request this year."**

This ensures the **one-laptop-request-per-employee-per-year** policy is enforced automatically at the backend level.





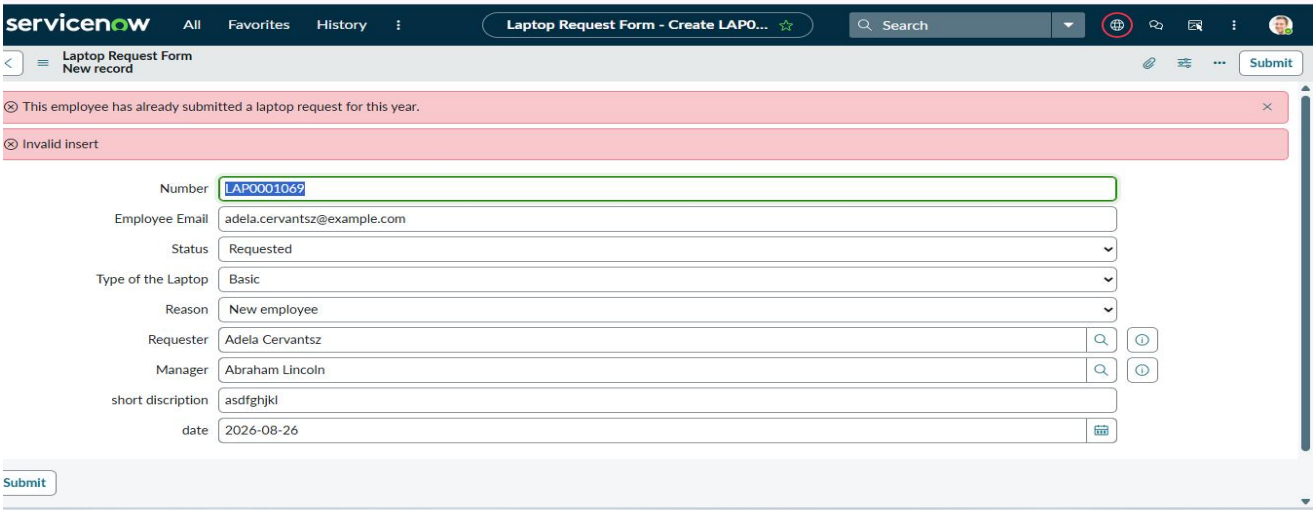
Duplicate Request Validation

When an employee tries to submit a **second laptop request in the same calendar year**, the Business Rule checks the existing requests for that employee.

If a duplicate request is found, the system **blocks the submission** and displays an error message:

"You have already submitted a laptop request this year. Duplicate requests are not allowed."

Flow:
Submit Request → Check Existing Request → Duplicate Found → Error Message → Request Not Created



Role-Based Access Control

Purpose:
To control access to the Laptop Request Management application based on user roles.

Navigation:
All → User Administration → Roles → New

Create the following two roles:

Role Name Purpose

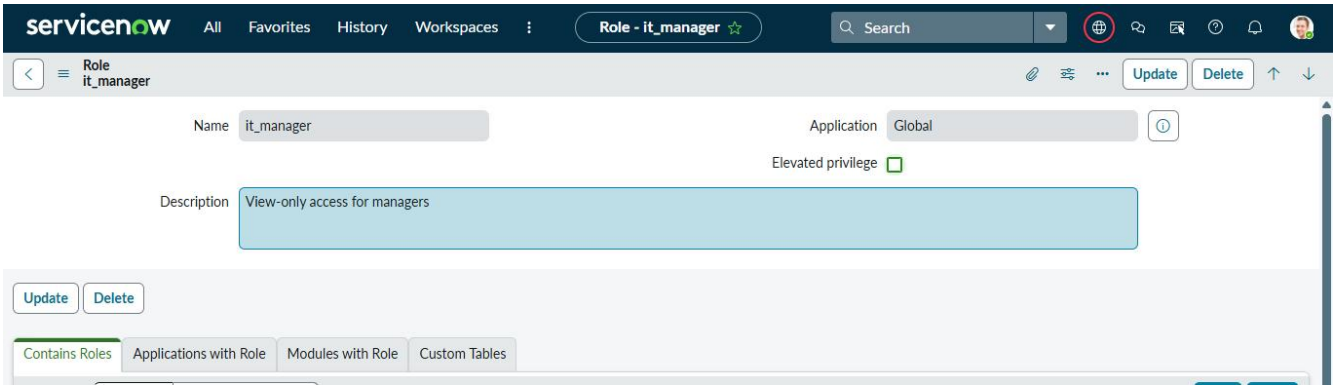
it_manager View-only access to laptop requests for IT managers

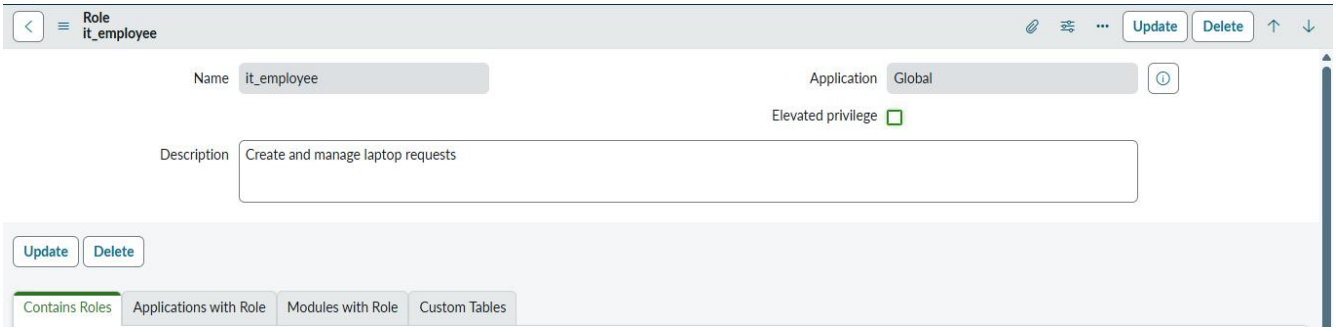
it_employee Create and manage laptop requests for IT employees

Short Explanation:

- **it_employee:** Can create and manage their laptop requests.
- **it_manager:** Can view laptop requests for monitoring and approval purposes.

Access Flow:
User Login → Role Verification → Access Granted Based on Role → Application/Table Access

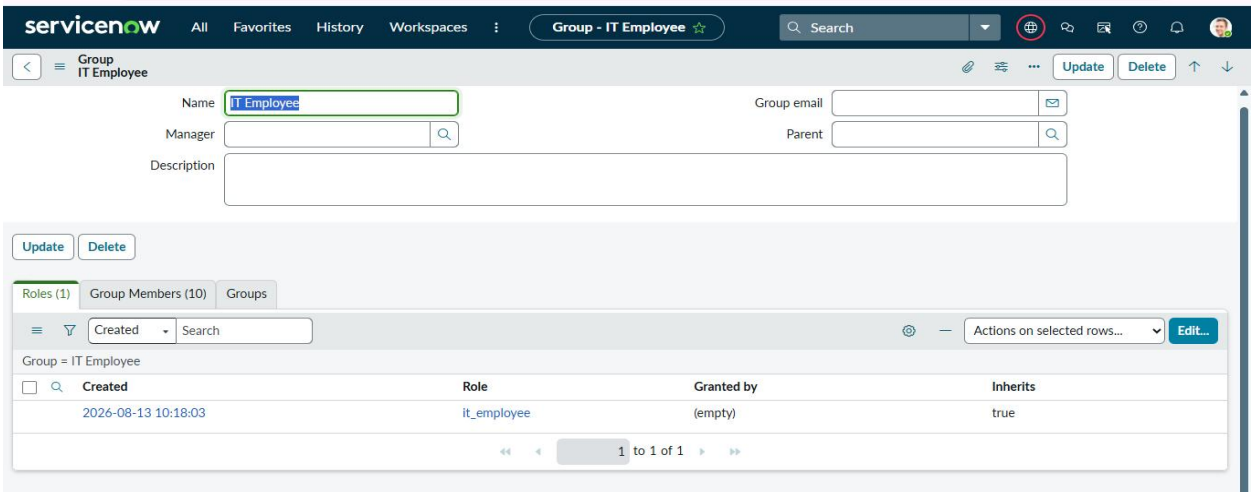




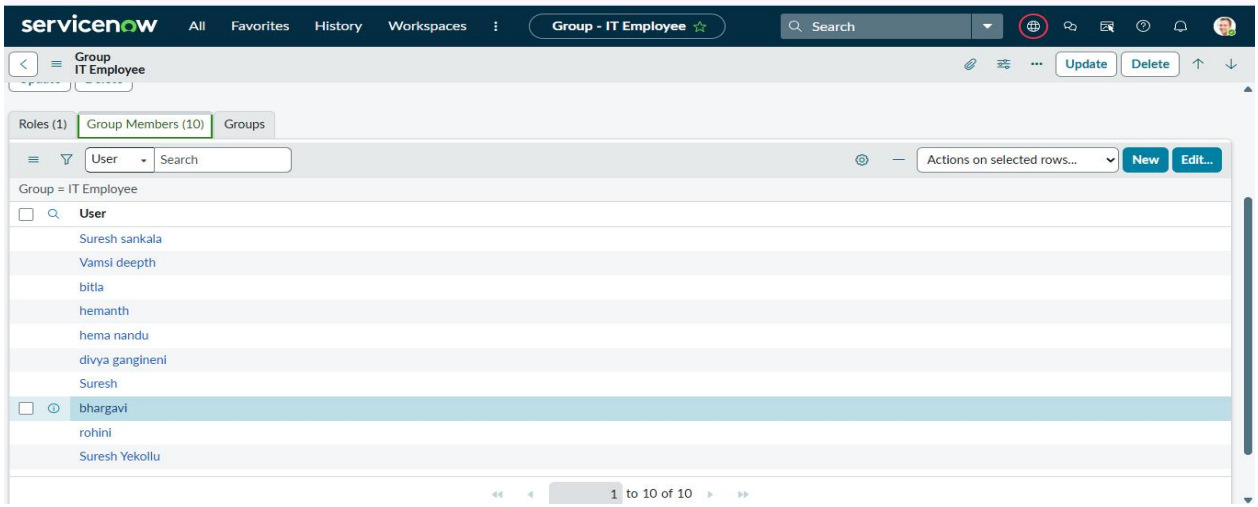
To manage access efficiently using groups.

- On All menus >> Search groups under users and groups select it.
- And create two new groups with given names and add members to those groups which are already created.

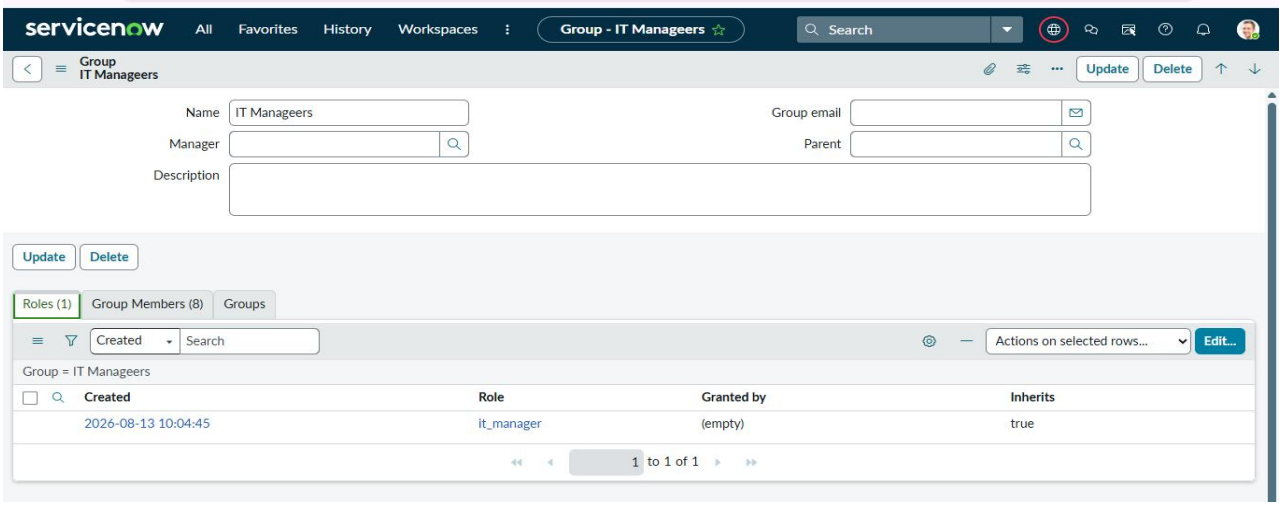
- Now add two roles to the two groups: manager to It manager role and Employee to It employee role.

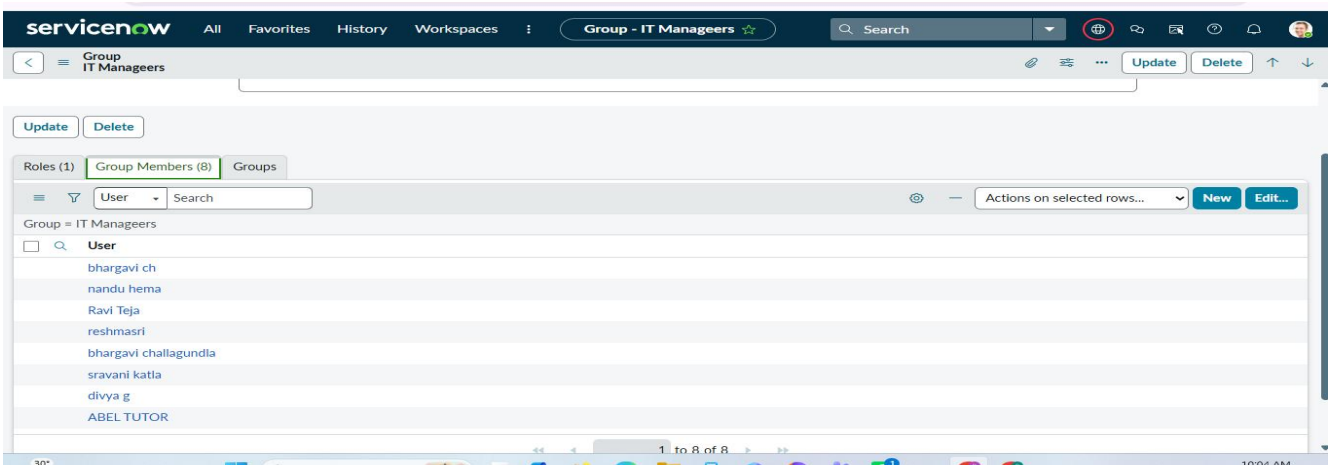


Also add some group members.



Next create another group and add some group members.





ACL creation

Purpose:
To restrict access to the **Laptop Request Form** based on user roles.

1. Manager Access – Read Only

- **Role:** it_manager
- **ACL:** Read
- **Table:** Laptop Request Form
- **Result:** Managers can **view** requests but cannot create or modify them.

2. Employee Access – Full Management

- **Role:** it_employee
- **ACLs:** Read, Create, Write
- **Table:** Laptop Request Form
- **Result:** Employees can **create, view, and edit** laptop requests.

Navigation in ServiceNow

System Security → Access Control (ACL)

1. Click **New**.
2. Select **Type:** record.
3. Select **Operation:** Read.
4. Select **Table:** Laptop Request Form.
5. Add role: it_manager.

6. Click **Submit**.

Then create three ACLs for employees:

System Security → Access Control (ACL) → New

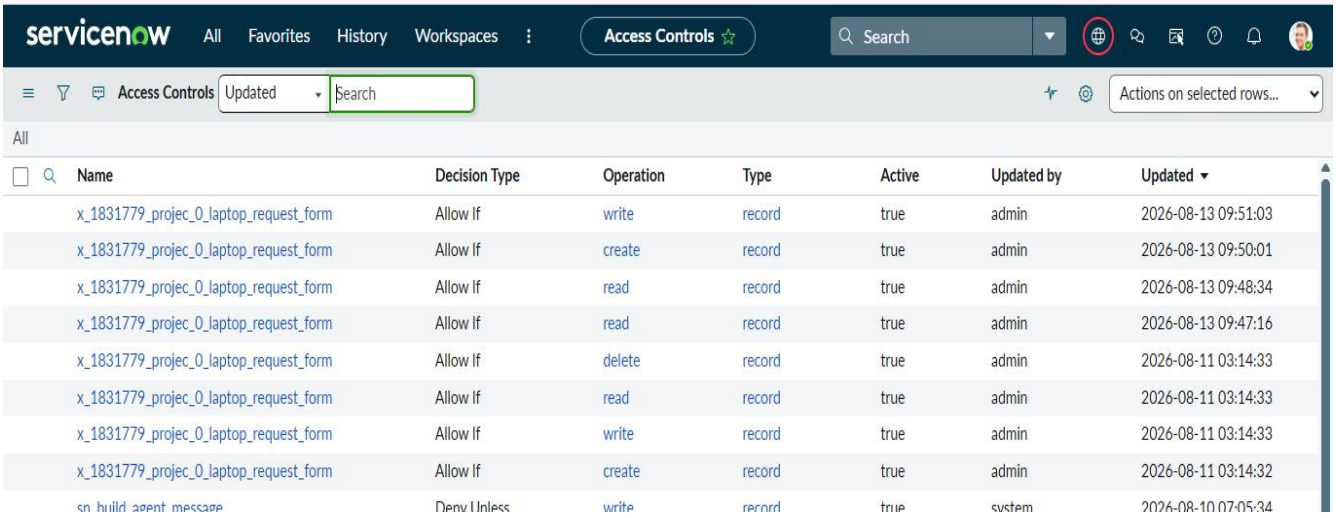
- Read → Laptop Request Form → it_employee
- Create → Laptop Request Form → it_employee
- Write → Laptop Request Form → it_employee

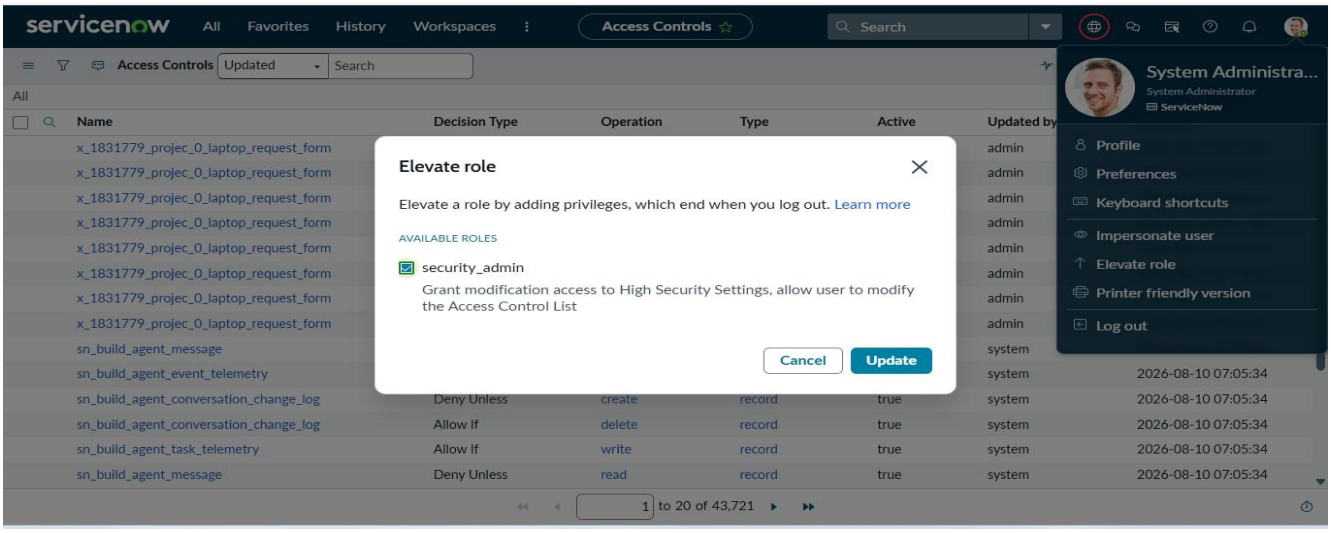
ACL Design

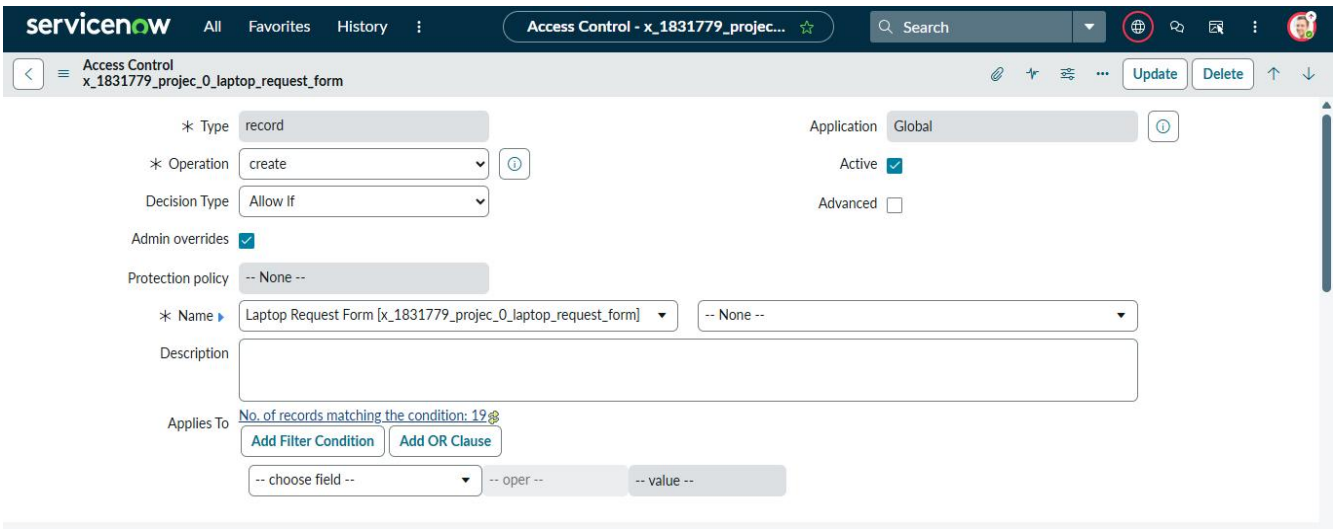
Use **Table-level ACLs** with the format:

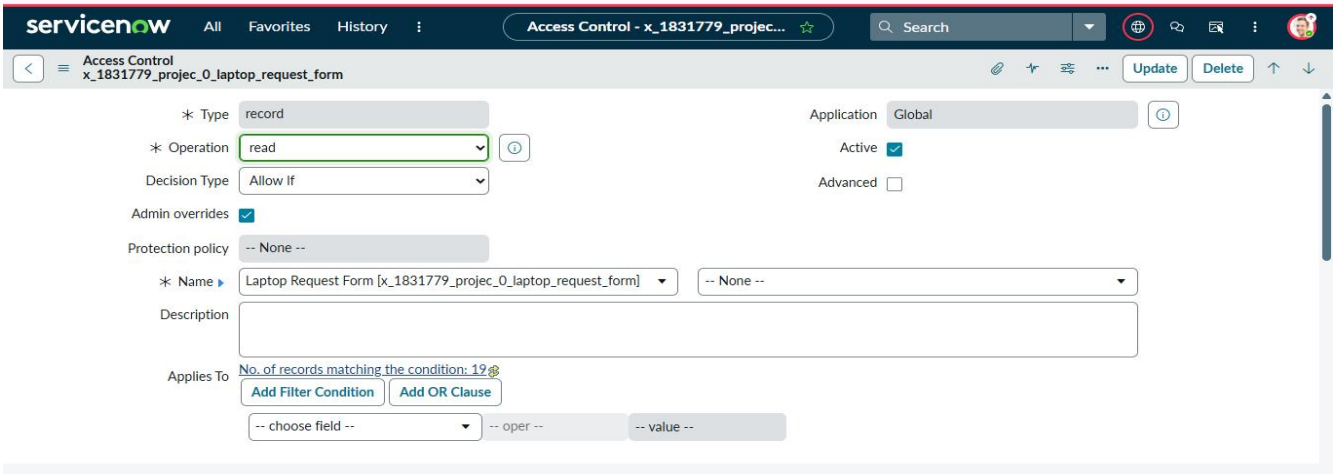
tablename.none

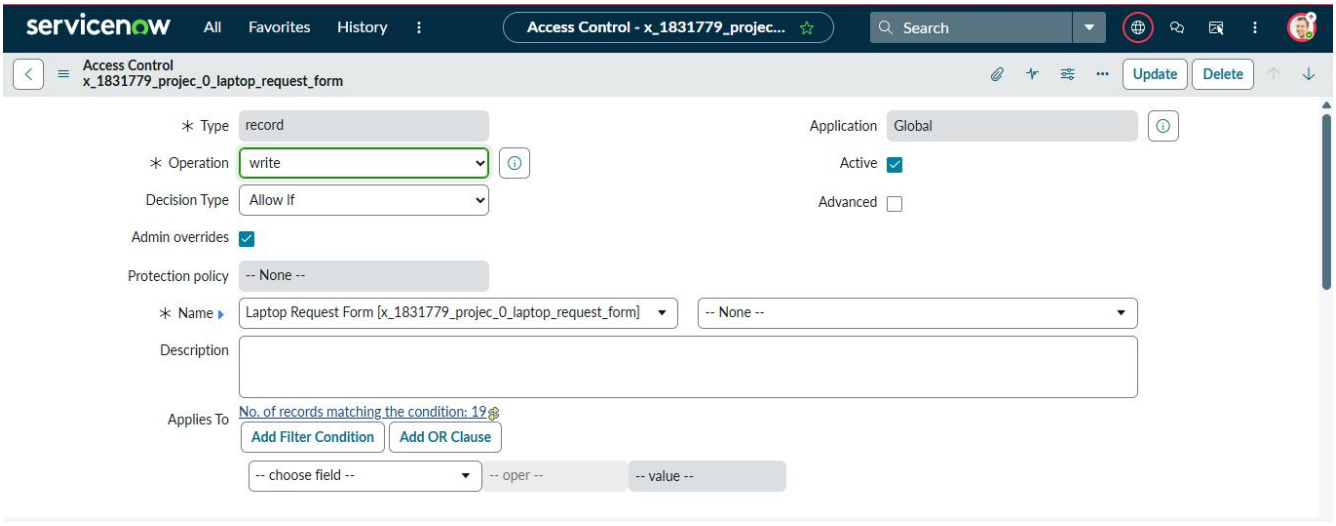
This provides consistent access control across the entire Laptop Request table.











Opened HR Onboarding Application Update Set

Verified inclusion of:

Custom Application (Project Application)

Laptop Request Form table & fields

Record Producer

Business Rules

ACLs

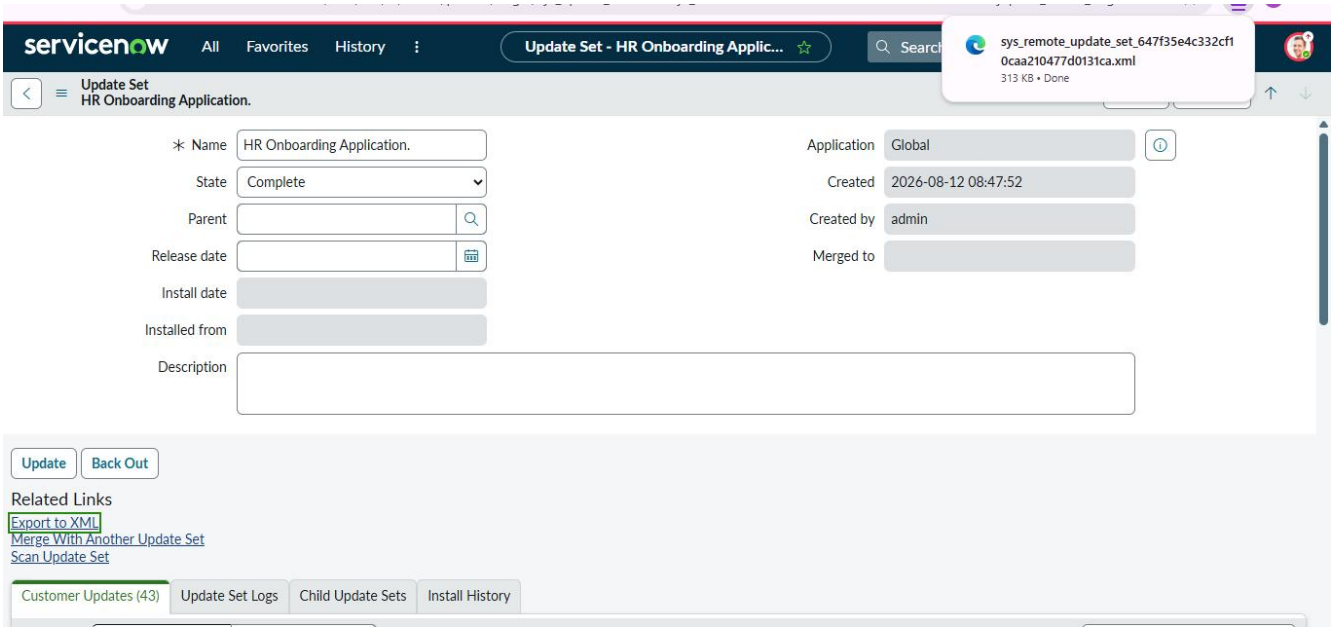
Flow Designer approval workflow

Email notifications

Performed Update Set Preview

Resolved conflicts.

Committed update set to target instance after changing state to complete and export to xml.



Conclusion

The **Laptop Request Management System** was successfully developed in ServiceNow to automate and streamline the laptop request process. The system manages request creation, manager approval, status tracking, notifications, duplicate-request validation, and role-based access control. Employees can create and manage their requests, while managers have controlled read-only access. Overall, the project improves **security, efficiency, transparency, and request management** within the organization.